

■ ELECTRONIC CONTROL SYSTEM

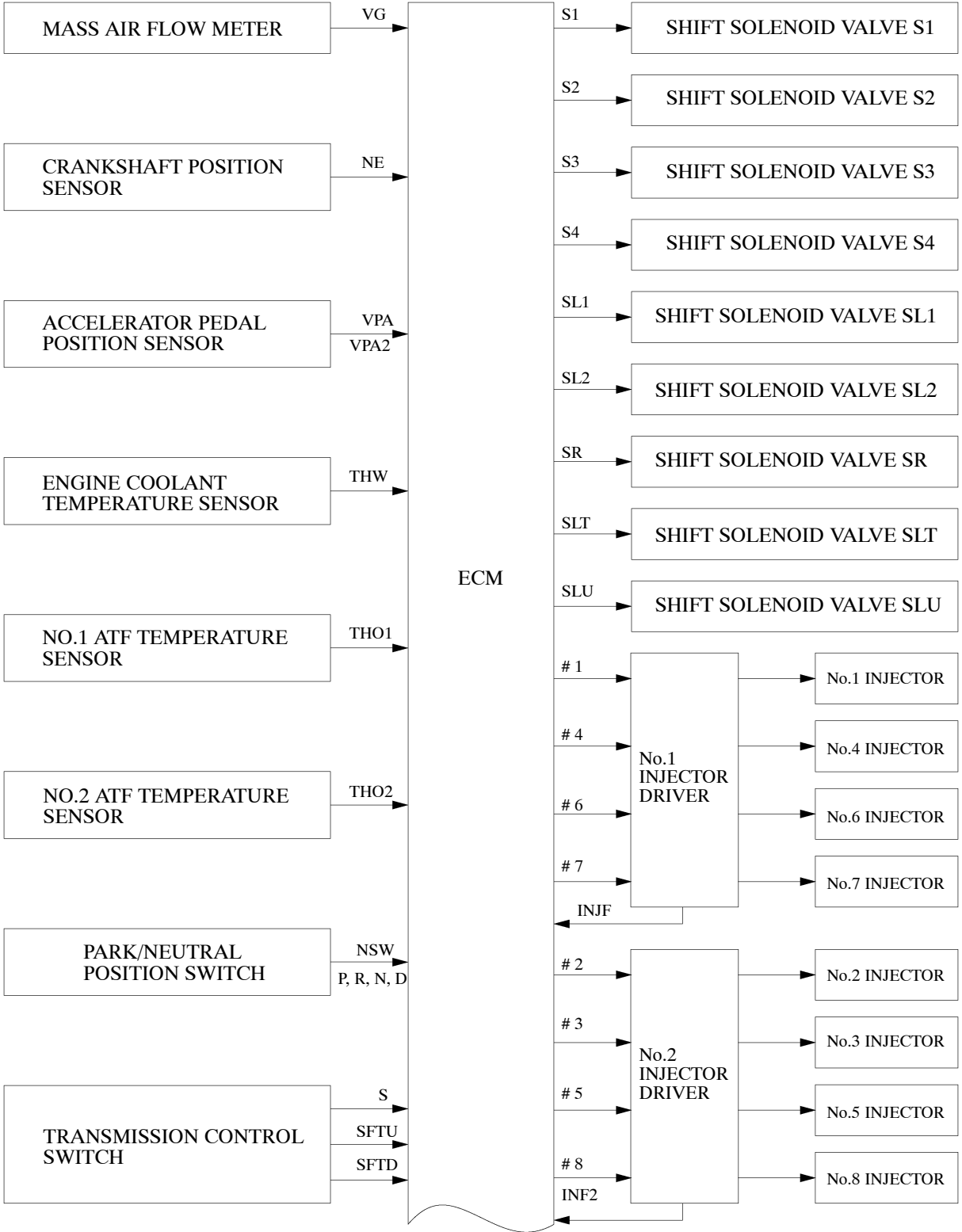
1. General

The electronic control system of the AB60F automatic transmissions consists of the control functions listed below.

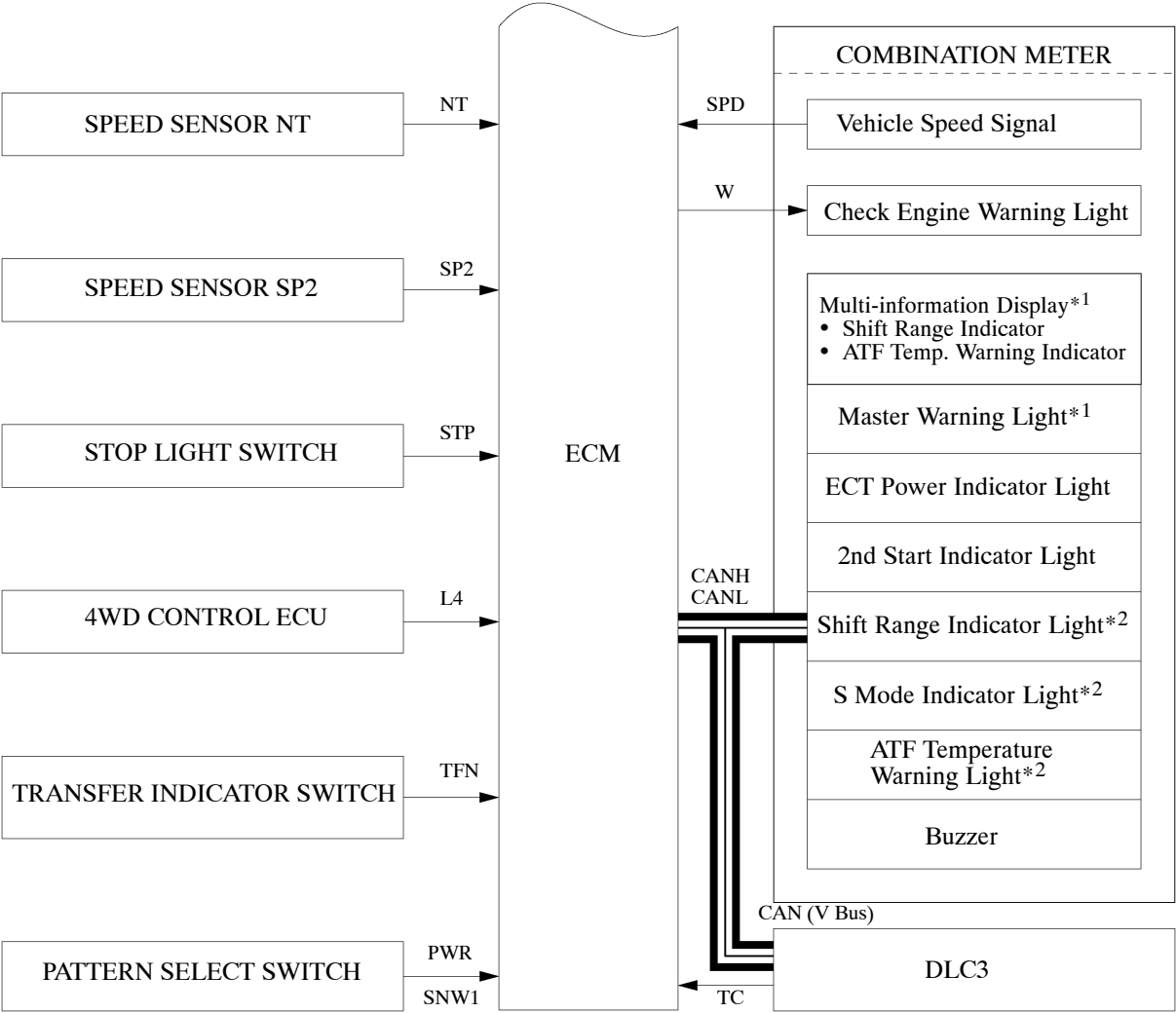
System	Function
Shift Timing Control	The ECM sends current to shift solenoid valves S1, S2, S3, S4 or SR based on signals from various sensors, in order to shift the gears.
Clutch Pressure Control (See Page CH-71)	- Controls the pressure that is applied directly to B₂ brake and C₃ clutch by actuating the linear solenoid valves SL1 and SL2 in accordance with the ECM signals. - The shift solenoid valve SLT and SL1 minutely controls the clutch pressure in accordance with the engine output and driving conditions.
Line Pressure Optimal Control (See Page CH-72)	Actuates the shift solenoid valve SLT to control the line pressure in accordance with information from the ECM and the operating conditions of the transmission.
Engine Torque Control	Reduces the injection quantity temporarily to improve shift feeling while upshifts or downshifts occur.
Lock-up Timing Control (See Page CH-72)	The ECM sends current to the shift solenoid valve SLU based on signals from various sensors and engages or disengages the lock-up clutch.
Flex Lock-up Clutch Control (See Page CH-73)	Controls the shift solenoid valve SLU, provides an intermediate mode for when the lock-up clutch is between ON and OFF, increasing the operating range of the lock-up clutch to improve fuel economy.
“N” to “D” Squat Control	When the shift lever is shifted from “N” to “D” position, 2nd gear is temporarily engaged before 1st to reduce vehicle squat.
“R” to “D” Squat Control	When the shift lever is shifted from “R” to “D” position, 3rd gear is temporarily engaged before 1st to reduce vehicle squat.
2nd Start Control	Enabling the vehicle to take off in the 2nd gear and thus make it easy to take off snowy, sandy or muddy terrain.
Coast Downshift Control (See Page CH-74)	To prevent engine speed from decreasing and thereby maintain fuel cut, the ECM performs downshifts before fuel cut ends.
AI (Artificial Intelligence) -SHIFT Control (See Page CH-75)	Based on the signals from various sensors, the ECM determines the road conditions and the intention of the driver. While the pattern select switch is in POWER, the shift pattern is biased toward sporty driving. Thus, an appropriate shift pattern is automatically determined, thus improving drivability.
Multi-mode Automatic Transmission (See Page CH-77)	The ECM appropriately controls the automatic transmission in accordance with the range position selected while the shift lever is in the S mode position.
Diagnosis (See Page CH-79)	When the ECM detects a malfunction, the ECM records the malfunction and memorizes the information that relates to the fault.
Fail-safe (See Page CH-79)	If a malfunction is detected in the sensors or solenoids, the ECM effects fail-safe control to prevent the vehicle's drivability from being affected significantly.

2. Construction

The configurations of the electronic control system for the AB60F automatic transmission is as shown in the following chart.



(Continued)

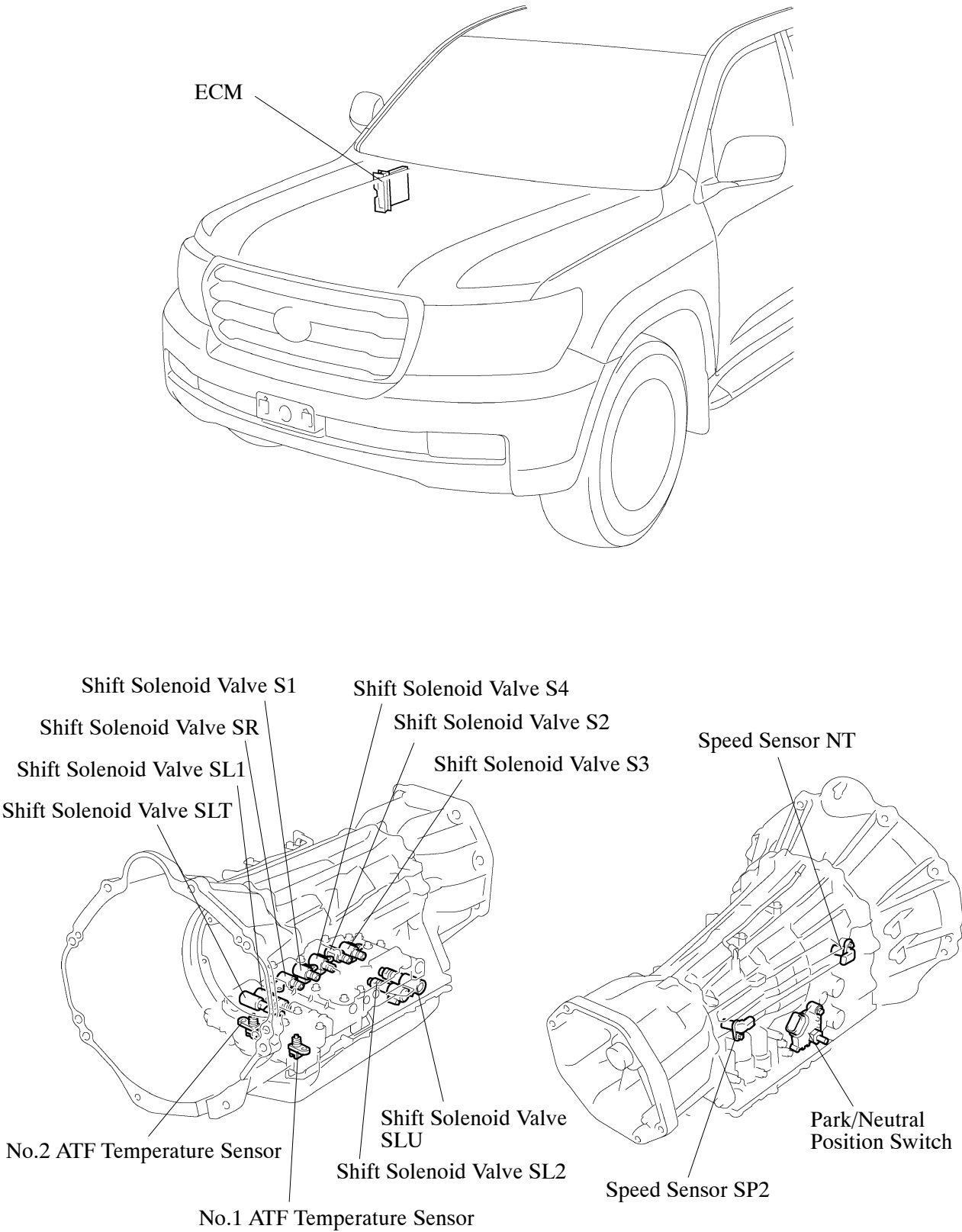


081CH63S

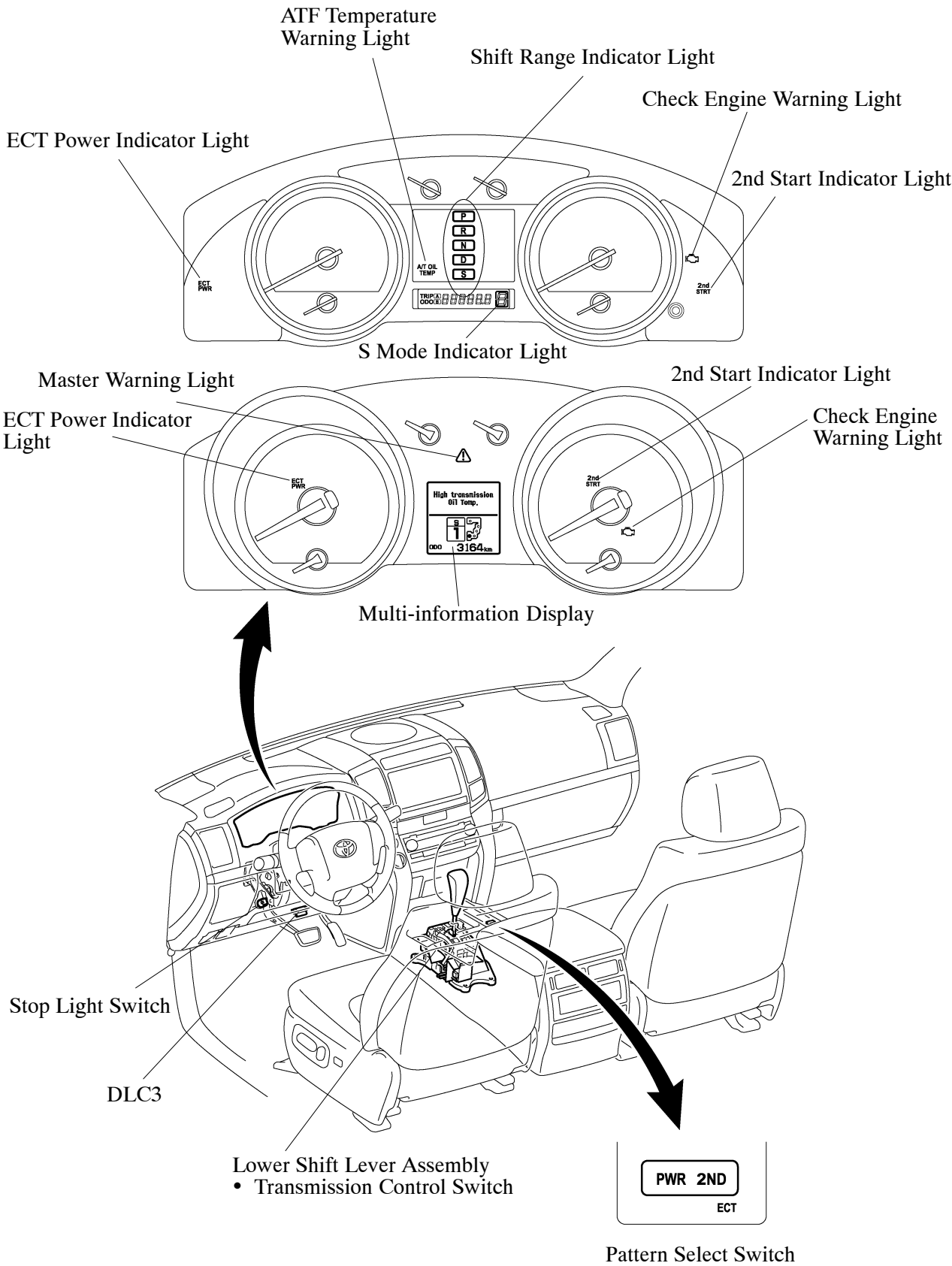
*1: Only for models with optitron type combination meter

*2: Only for models with analog type combination meter

3. Layout of Main Components



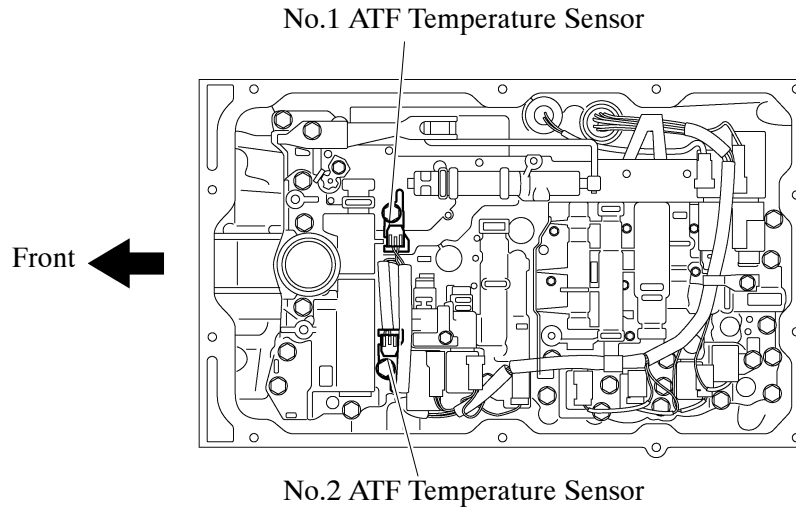
CH



4. Construction and Operation of Main Components

No.1 and No.2 ATF Temperature Sensor

- No.1 ATF temperature sensor is used for hydraulic pressure control. This sensor is used for revision of the pressure that is used to apply clutches and brakes in the transmission. This helps to ensure smooth shift quality.
- No.2 ATF temperature sensor is used as a basis for modifying the ECT shift timing control when the ATF temperature is high. It is also used for the ATF temperature warning light.



04E1CH17C

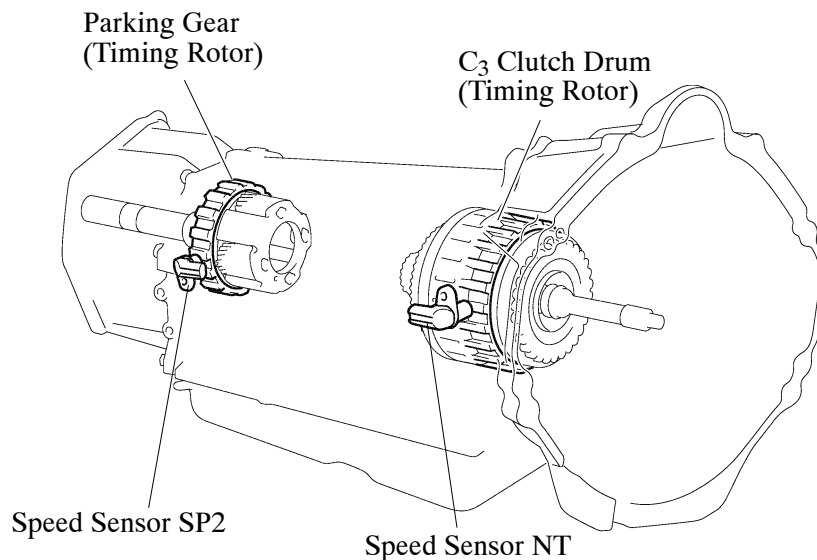
CH

Input Speed Sensor and Output Speed Sensor

The AB60F automatic transmissions use a speed sensor NT (for NT signal) and speed sensor SP2 (for SP2 signal). Thus, the ECM can detect the timing of the shifting of the gears and appropriately control the engine torque and hydraulic pressure in response to various conditions.

These speed sensors are the pick-up coil type.

- The speed sensor NT detects the input speed of the transmission. The clutch drum is used as the timing rotor for this sensor.
- The speed sensor SP2 detects the speed of the output shaft. The parking gear on the rear planetary gear is used as the timing rotor for this sensor.

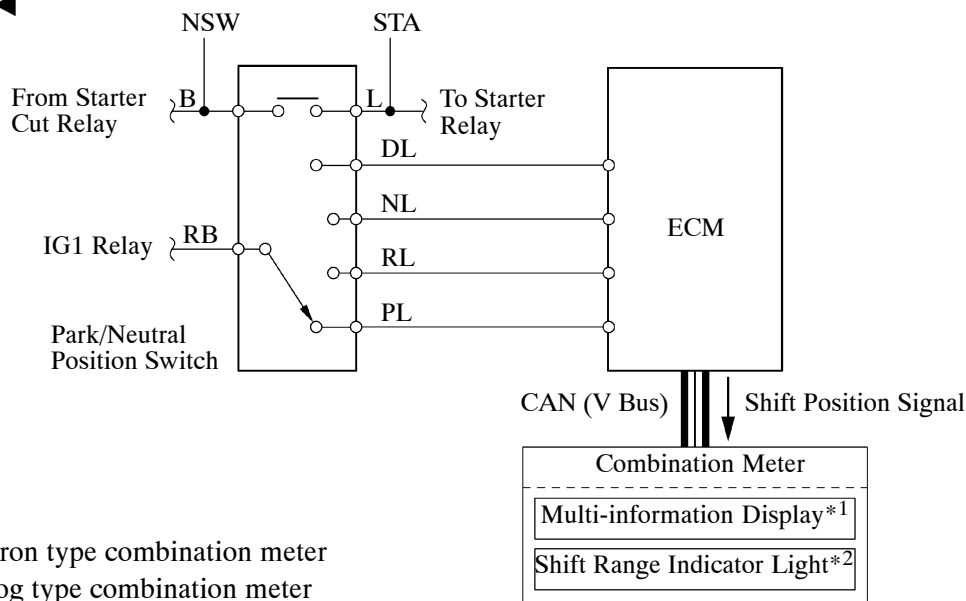


081CH42Z

Park/Neutral Position Switch

The park/neutral position switch sends the P, R, N, D and NSW position signals to the ECM. It also sends signals for the shift position indicator light (P, R, N, and D).

► Wiring Diagram ◀



*1: Models with optitron type combination meter

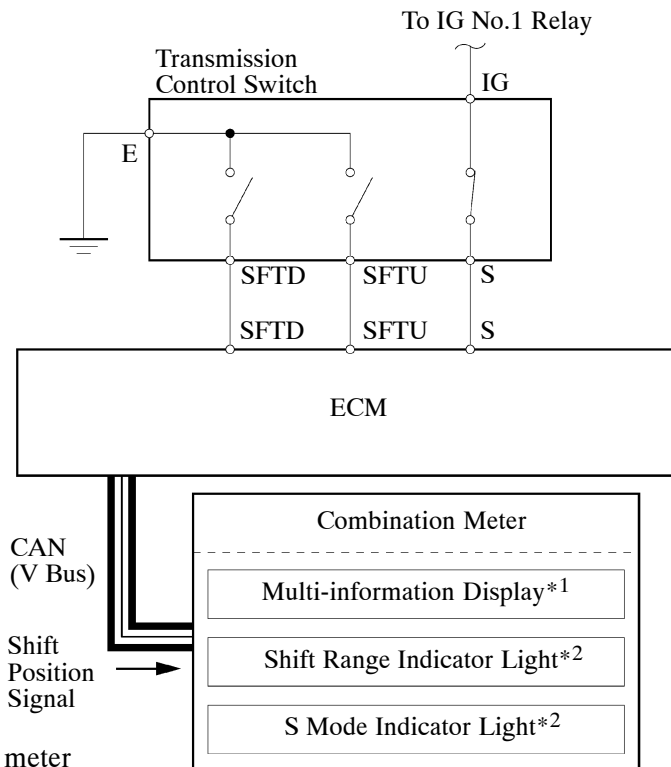
*2: Models with analog type combination meter

081CH28S

Transmission Control Switch

- The transmission control switch is installed inside the shift lever assembly to detect the shift lever position and to inform the ECM. The ECM turns on the shift position indicator light and S mode indicator light.
- The transmission control switch detects whether the shift lever is in the D position or in the S mode position, and detects the operating conditions of the shift lever (“+” position or “-” position) if the S mode is selected, and sends signals to the ECM. At this time, the ECM turns on the shift range indicator light for the selected range.

► Wiring Diagram ◀



*1: Models with optitron type combination meter

*2: Models with analog type combination meter

081CH60S

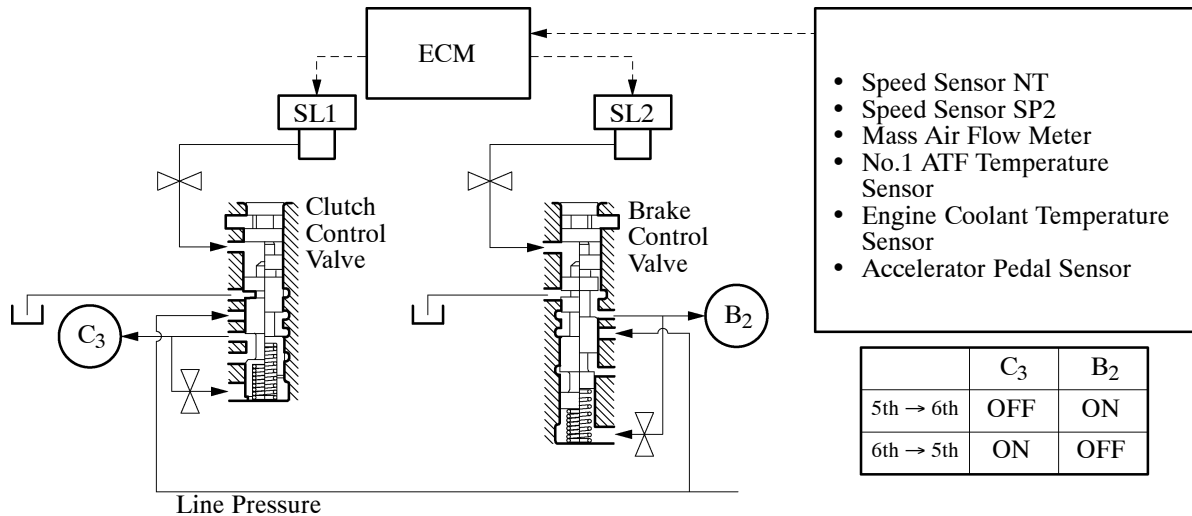
5. Clutch Pressure Control

Clutch to Clutch Pressure Control

This control is used for shifting from 5th to 6th gear and from 6th to 5th gear.

The ECM actuates shift solenoid valves SL1 and SL2 in accordance with various signals. The output from these solenoid valves acts directly on control valves B₂ and C₃ in order to regulate the line pressure that acts on the B₂ brake and C₃ clutch.

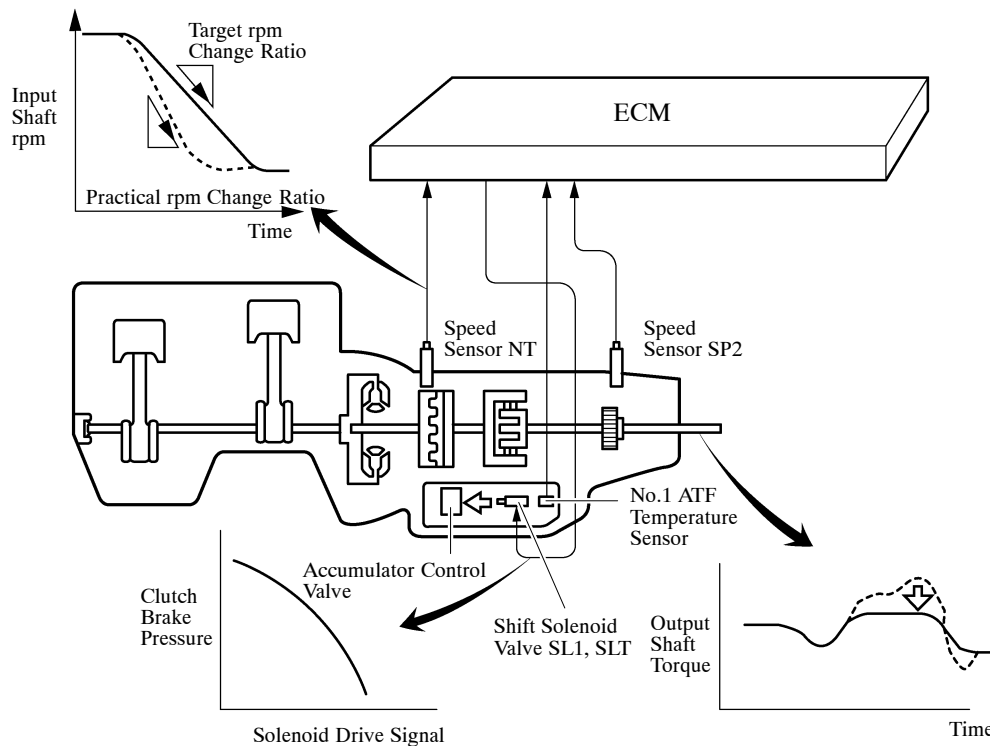
As a result, high response and excellent shift characteristics have been realized.



04E1CH07C

Clutch Pressure Optimal Control

The ECM monitors the signals from various types of sensors, such as the input speed sensor, allowing shift solenoid valves SLT and SL1 to minutely control the clutch pressure in accordance with engine output and driving conditions. As a result, smooth shift characteristics are realized.

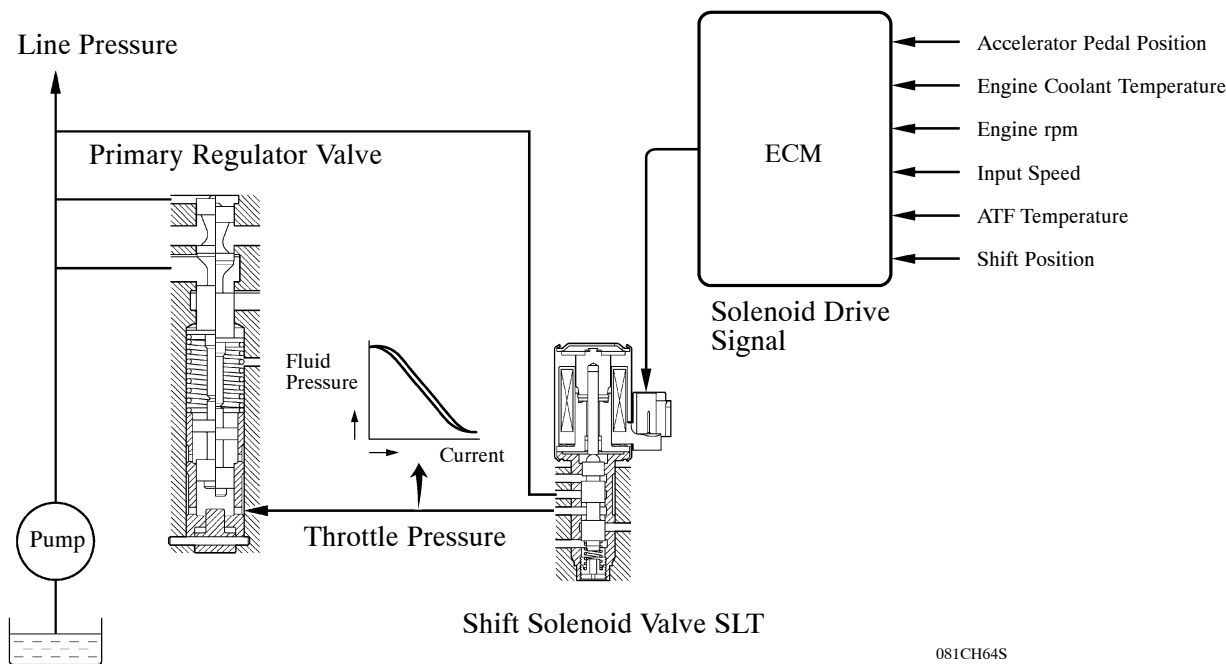


279CH109

6. Line Pressure Optimal Control

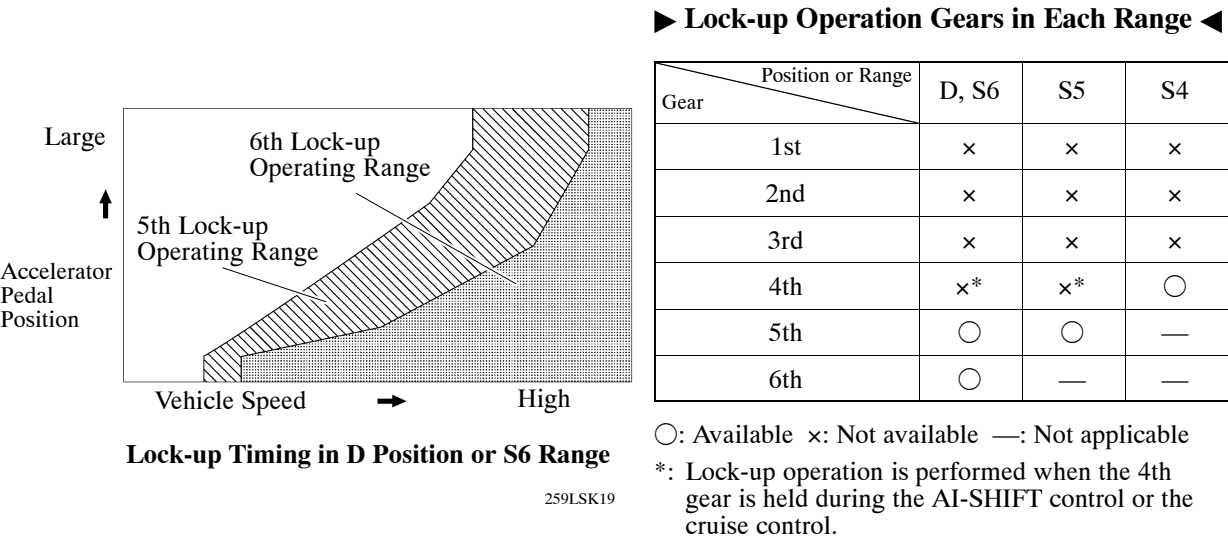
Through the use of the shift solenoid valve SLT, the line pressure is optimally controlled in accordance with the engine torque information, as well as with the internal operating conditions of the torque converter clutch and the transmission.

Accordingly, the line pressure can be controlled minutely in accordance with the engine output, traveling condition, and ATF temperature, thus realizing smooth shift characteristics and optimizing the workload of the oil pump (reducing unnecessary parasitic losses).



7. Lock-up Timing Control

The ECM operates the lock-up timing control in order to improve the fuel consumption performance while in top gear with the shift lever in the S4 or S5 range, and in 5th or 6th gear with the shift lever in the S6 range or D position.

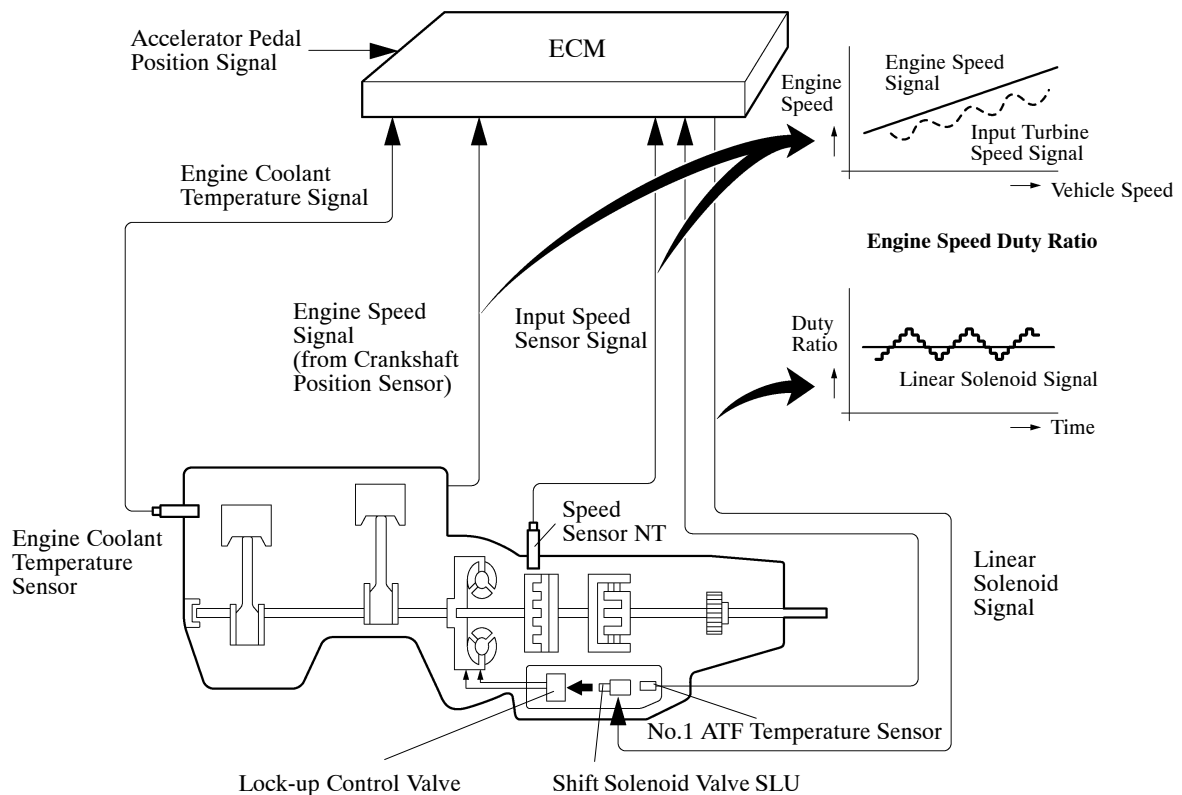


8. Flex Lock-up Clutch Control

In the low-to-mid-speed range, this flex lock-up clutch control regulates the shift solenoid valve SLU to provide an intermediate mode between the ON/OFF operation of the lock-up clutch in order to improve the energy transmitting efficiency in this range.

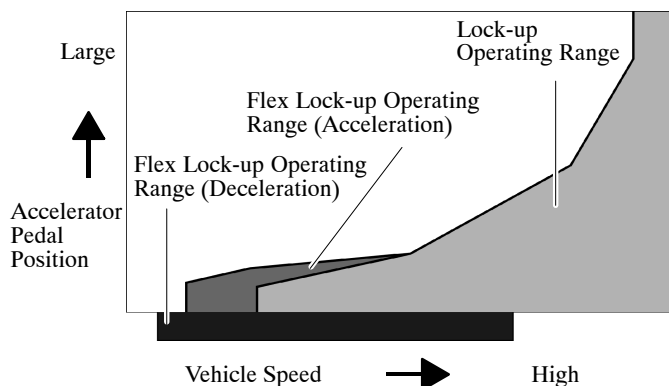
As a result, the operating range of the lock-up clutch has been increased and fuel economy has been improved. The flex lock-up clutch control operates in the 3rd, 4th, 5th and 6th gears in the D position and S6 range, 3rd, 4th and 5th gears in the S5 range, 3rd and 4th gear in the S4 range.

- Even when the vehicle is decelerating (the accelerator pedal is released), the flex lock-up clutch control operates. Therefore, fuel-cut area of the engine has been expanded and fuel-economy has been improved.



081CH65S

► Flex Lock-up Operation Gears in Each Range ◀



Position or Range	D, S6	S5	S4
Gear			
1st	×	×	×
2nd	×	×	×
3rd	○	○	○
4th	○*	○*	○*
5th	○*	○*	—
6th	○*	—	—

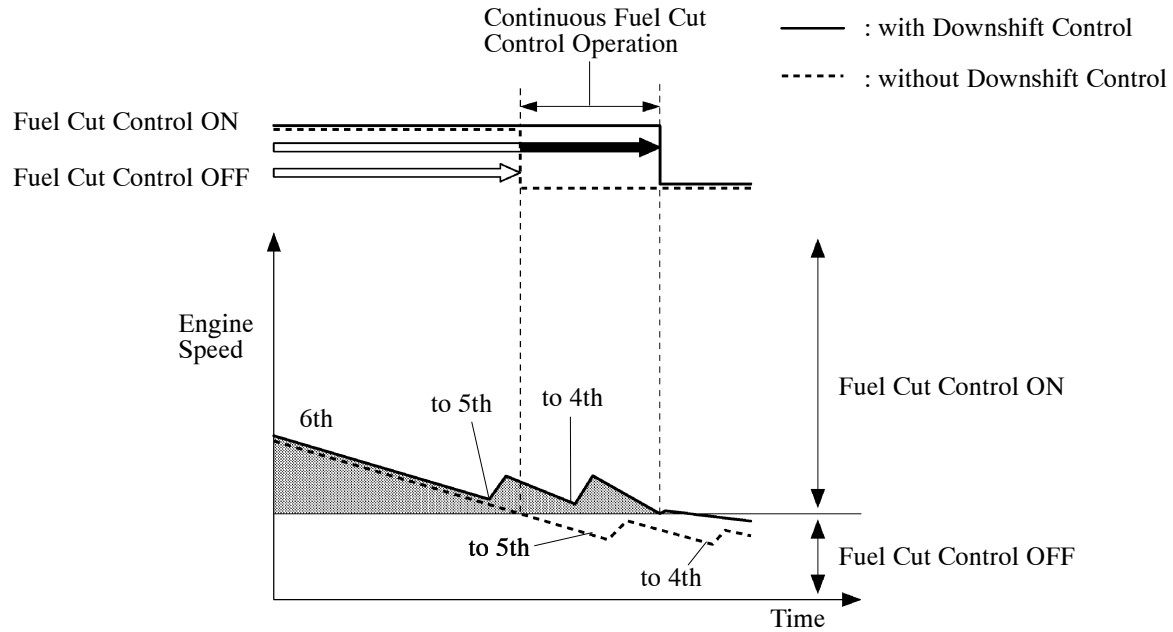
○: Available ×: Not available —: Not applicable
 *: Flex Lock-up also operates, when the vehicle is decelerating.

0140CH141C

9. Coast Downshift Control

As a result of coast downshift control, downshifts are performed to maintain sufficient engine speed to avoid ending fuel cut control. Thus, fuel cut time is extended and fuel economy is achieved.

- In this control, the transmission downshifts from 6th to 5th and then 5th to 4th before fuel cut control ends when the vehicle is decelerated in the 6th gear, so that fuel cut control continues operating.



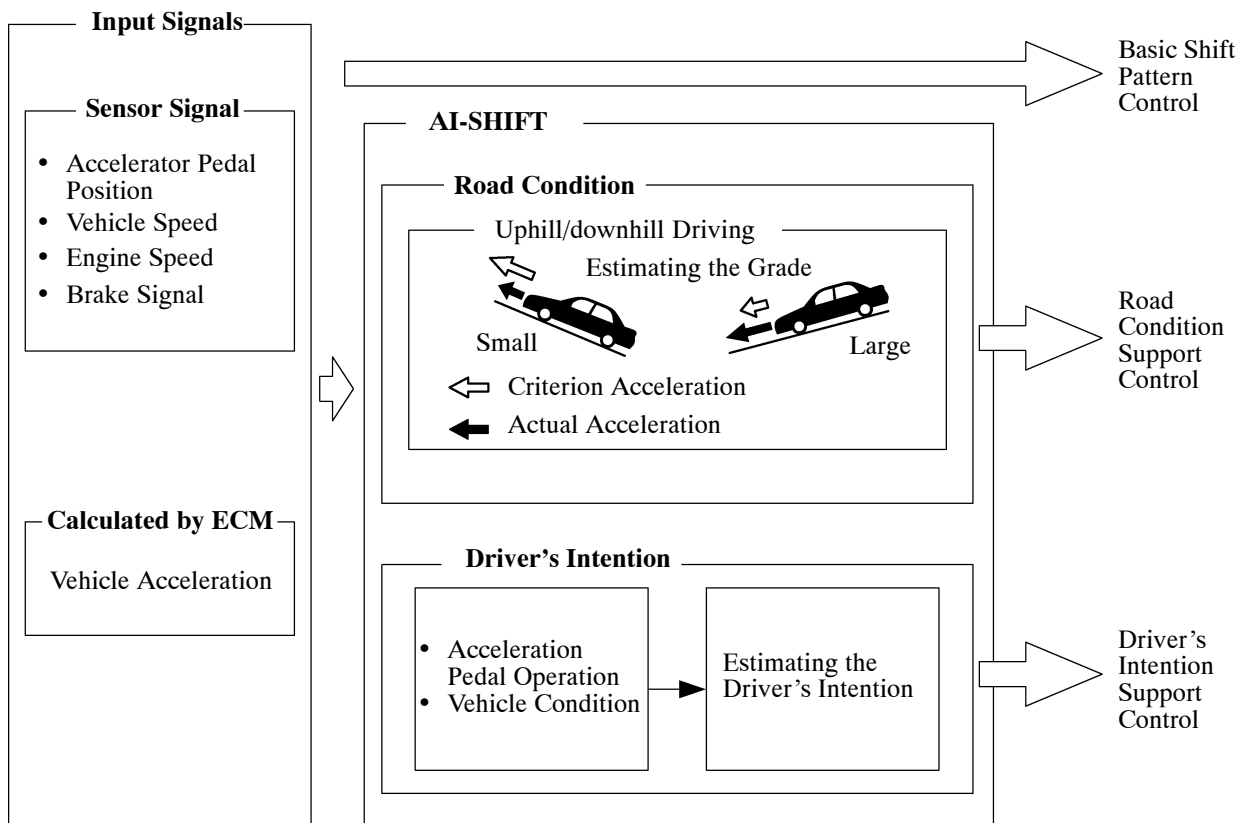
04E1CH12C

10. AI (Artificial Intelligence) - SHIFT Control

General

The AI-SHIFT control includes a road condition support control and a driver's intention support control. The AI-SHIFT control determines optimal transmission control based on input signals and automatically changes the shift pattern. As a result, a high caliber of transmission operation is achieved.

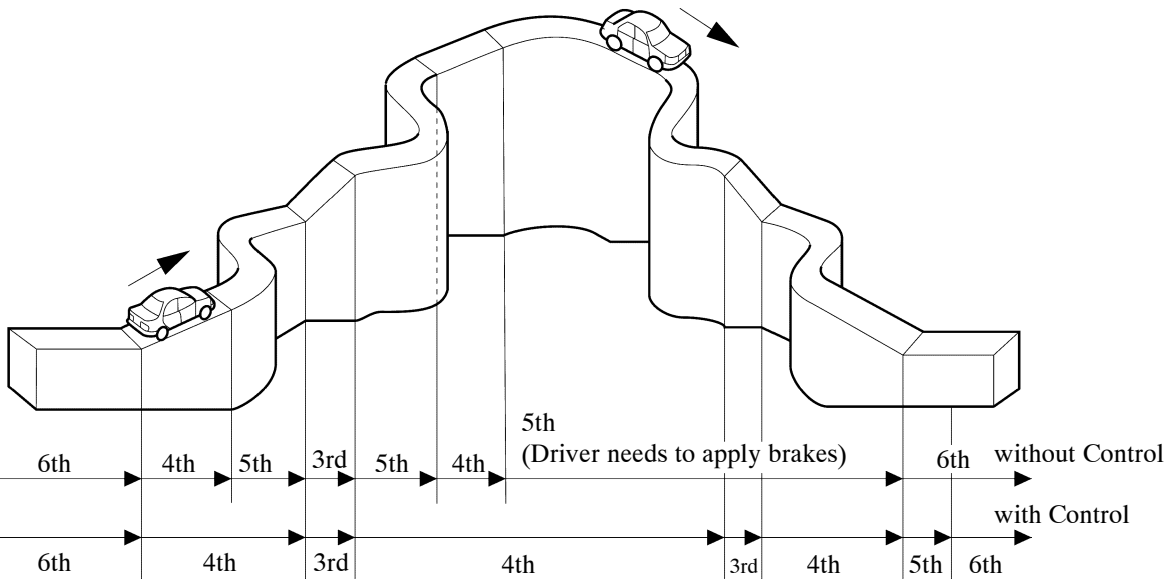
- AI-SHIFT control is effect only with the shift lever in the D position, based on the accelerator and brake operation data. AI-SHIFT control will be canceled when the driver selects the S mode.



Road Condition Support Control

Under road condition support control, the ECM determines accelerator pedal position and the vehicle speed whether the vehicle is being driven uphill or downhill.

- To achieve an optimal drive force while driving uphill, this control prevents the transmission from up-shifting to 4th, 5th or 6th gear.
- To achieve an optimal engine braking effect while driving downhill, this control automatically downshifts the transmission to 5th, 4th or 3rd gear.



040SC13C

Driver's Intention Support Control

Driver's intention support control estimates the driver's intention based on the accelerator operation and vehicle condition and selects a shift pattern that is well-suited to each driver.

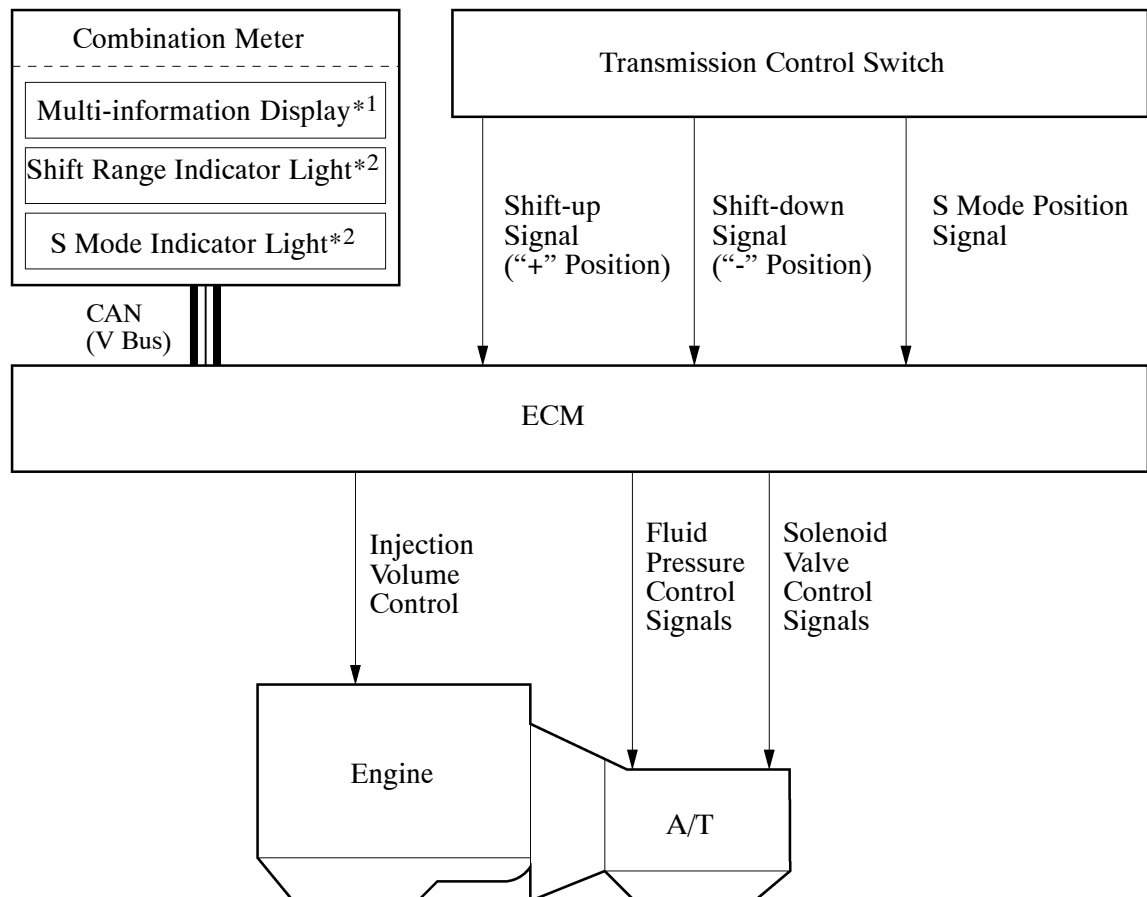
Control	Operation	Operate Gear	Pattern Select Switch
Sudden Accelerator Pedal Release Control	When the driver releases the accelerator quickly, this control makes it easy for the transmission to hold the gear, which improves engine braking force and re-acceleration response.	3rd to 5th	ECT POWER
Sudden Deceleration Downshift Control	When the driver decelerates the vehicle suddenly, this control downshifts rapidly, which improves engine braking force and re-acceleration response.	4th to 6th	ECT POWER

11. Multi-mode Automatic Transmission

General

A multi-mode automatic transmission is designed to allow the driver to switch the gear ranges (a multi-mode transmission is not for manually selecting single gears). After shifting the shift lever to the S mode position and moving the shift lever to the “+” position or to the “-” position, the driver can select the desired shift range. Thus, the driver is able to shift gears with a manual-like feel.

► System Diagram ◀



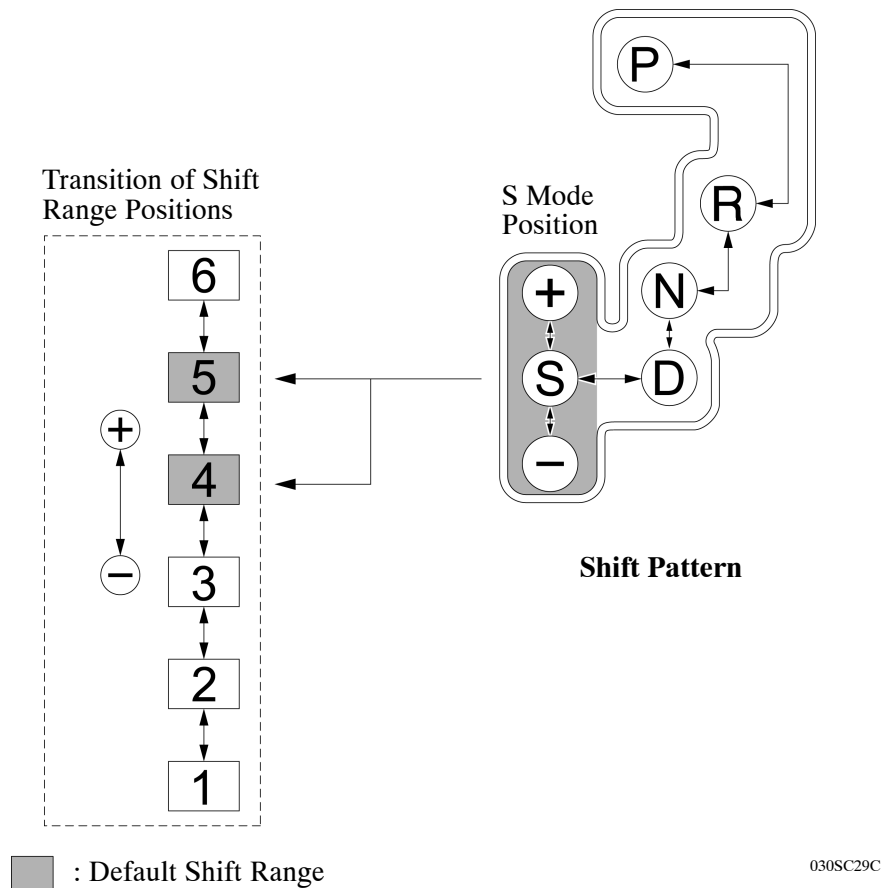
CH

*1: Models with optitron type combination meter

*2: Models with analog type combination meter

Operation

- The driver selects the S mode position by engaging the shift lever. At this time, the 4th or 5th range is selected according to the vehicle speed. (When the driver selects the S mode during AI-SHIFT control, the current gear will be selected as the shift range.) Then, the shift range positions change one at a time, as the driver moves the shift lever to the front (“+” position) or to the rear (“-” position).
- Under this control, the ECM effects optimal shift control within the usable gear range that the driver has selected. As with an ordinary automatic transmission, it shifts to the 1st gear when the vehicle is stopped.
- Holding the transmission control switch in the “+” position with the shift lever in the S mode position will change the shift range to the 6th range regardless of range position (1st to 5th).
- When the shift lever is in the S mode position, the S mode indicator light in the combination meter illuminates. The shift range indicator light indicates the state of the shift range position that the driver has selected.



► Usable Gear Chart ◀

Shift Range	Shift Range Indicator Light Indication	Usable Gears
6th	6	6th ↔ 5th ↔ 4th ↔ 3rd ↔ 2nd ↔ 1st
5th	5	5th ↔ 4th ↔ 3rd ↔ 2nd ↔ 1st
4th	4	4th ↔ 3rd ↔ 2nd ↔ 1st
3rd	3	3rd ↔ 2nd ↔ 1st
2nd	2	2nd ↔ 1st
1st	1	1st

12. Diagnosis

- When the ECM detects a malfunction, the ECM records the malfunction and memorizes the information related to the fault. Furthermore, the check engine warning light in the combination meter illuminates or blinks to inform the driver.
- At the same time, the DTC (Diagnostic Trouble Code) is stored in memory. The DTC can be read by connecting an intelligent tester II to the DLC3.

For details, see the Land Cruiser (Station Wagon) Repair Manual (Pub. No. RM0810E).

13. Fail-safe

The fail-safe function minimizes the loss of operability when an abnormality occurs in a sensor or solenoid.

► Fail-safe Control List ◀

Malfunction Part	Function
Speed Sensor NT	When the speed sensor NT malfunctions, shift control is effected using the information from the speed sensor SP2 signal. During a speed sensor NT malfunction, up-shift to the 5th, 6th, AI-SHIFT and flex lock-up clutch control are prohibited.
Speed Sensor SP2	When the speed sensor SP2 malfunctions, shift control is effected using the information from the speed sensor NT signal. When the speed sensor SP2 malfunctions, up-shift to the 5th, 6th, AI-SHIFT and flex lock-up clutch control are prohibited.
No.1 ATF Temperature Sensor	When the No.1 ATF temperature sensor malfunctions, up-shift to the 5th, 6th and flex lock-up clutch control are prohibited.
Shift Solenoid Valve S1, S2, S3, S4 and SR	When a shift solenoid valve listed at left fails, the current to the failed solenoid valve is cut off. Shift control is changed to a fail-safe mode to shift gears using the normal shift solenoid valves to allow continued driving. Refer to the table on the next page for an operation example.
Shift Solenoid Valve SL1 and SL2	During a shift solenoid valve SL1 or SL2 malfunction, up-shift to the 5th and 6th and flex lock-up clutch control are prohibited.
Shift Solenoid Valve SLU	During a shift solenoid valve SLU malfunction, the current to the solenoid valve is stopped. Because this stops lock-up control and flex lock-up control, fuel economy decreases.
Shift Solenoid Valve SLT	During a shift solenoid valve SLT malfunction, the current to the solenoid valve is stopped. Because this stops line pressure optimal control, the shift shock increases. However, shifting is effected through normal clutch pressure control.

Position	Normal							
	Shift Solenoid							Gear
	S1	S2	S3	S4	SR	SL1	SL2	
D, S6	OFF	ON	ON	OFF	ON	OFF	ON	1st
	ON	ON	ON	OFF	ON	OFF	ON	2nd
	ON	OFF	ON	OFF	ON	OFF	ON	3rd
	ON	OFF	OFF	OFF	ON	OFF	ON	4th
	ON	OFF	OFF	ON	OFF	ON	OFF	5th
	ON	ON	OFF	ON	OFF	ON	OFF	6th
S5	OFF	ON	ON	OFF	ON	OFF	ON	1st
	ON	ON	ON	OFF	ON	OFF	ON	2nd
	ON	OFF	ON	OFF	ON	OFF	ON	3rd
	ON	OFF	OFF	OFF	ON	OFF	ON	4th
	ON	OFF	OFF	ON	OFF	ON	OFF	5th
S4	OFF	ON	ON	OFF	ON	OFF	ON	1st
	ON	ON	ON	OFF	ON	OFF	ON	2nd
	ON	OFF	ON	OFF	ON	OFF	ON	3rd
	ON	OFF	OFF	OFF	ON	OFF	ON	4th
S3	OFF	ON	ON	OFF	ON	OFF	ON	1st
	ON	ON	ON	OFF	ON	OFF	ON	2nd
	ON	OFF	ON	OFF	ON	OFF	OFF	3rd
S2	OFF	ON	ON	OFF	ON	OFF	ON	1st
	ON	ON	ON	ON	ON	OFF	OFF	2nd
S1	OFF	ON	ON	OFF	ON	OFF	OFF	1st

► Example ◀

Position	Shift Solenoid Valve S1 Malfunction							
	Shift Solenoid							Gear
	S1	S2	S3	S4	SR	SL1	SL2	
D, S6	×	ON	ON	OFF	ON	OFF	ON	1st
	×	ON → OFF	ON → OFF	OFF	ON	OFF	ON	1st → 4th
	×	OFF	ON → OFF	OFF	ON	OFF	ON	3rd → 4th
	×	OFF	OFF	OFF	ON	OFF	ON	4th
	×	OFF	OFF	ON	OFF	ON	OFF	5th
	×	ON → OFF	OFF	ON	OFF	ON	OFF	N → 5th
S5	×	ON	ON	OFF	ON	OFF	ON	1st
	×	ON → OFF	ON → OFF	OFF	ON	OFF	ON	1st → 4th
	×	OFF	ON → OFF	OFF	ON	OFF	ON	3rd → 4th
	×	OFF	OFF	OFF	ON	OFF	ON	4th
	×	ON → OFF	OFF	ON	OFF	ON	OFF	5th
S4	×	ON	ON	OFF	ON	OFF	ON	1st
	×	ON → OFF	ON → OFF	OFF	ON	OFF	ON	1st → 4th
	×	OFF	ON → OFF	OFF	ON	OFF	ON	3rd → 4th
	×	OFF	OFF	OFF	ON	OFF	ON	4th
S3	×	ON	ON	OFF	ON	OFF	ON	1st
	×	ON → OFF	ON → OFF	OFF	ON	OFF	ON	1st → 4th
	×	OFF	ON → OFF	OFF	ON	OFF	OFF → ON	3rd (E/B) → 4th
S2	×	ON	ON	OFF	ON	OFF	ON	1st
	×	ON → OFF	ON → OFF	ON	ON → OFF	OFF	OFF → ON	1st (E/B) → 2nd
S1	×	ON	ON	OFF	ON	OFF	OFF	1st (E/B)

E/B: Engine Braking